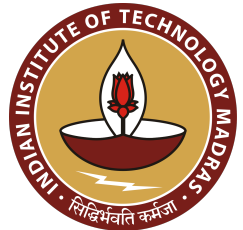


Differential Gene Expression Analysis

Nirav Bhatt

Wadhvani School of Artificial Intelligence (WSAI), IIT Madras



- RNA sequencing provides us with a snapshot of the biological activity

- RNA sequencing provides us with a snapshot of the biological activity
- But it isn't informative to just know the state of one cell

Introduction

- RNA sequencing provides us with a snapshot of the biological activity
- But it isn't informative to just know the state of one cell
- We need to compare it to other conditions

Introduction

- RNA sequencing provides us with a snapshot of the biological activity
- But it isn't informative to just know the state of one cell
- We need to compare it to other conditions



Example

Identifying genes having a statistically different expression levels in a diseased individual and healthy individual. Such genes are referred to as “dysregulated genes”.

- The conditions between which comparisons happen are called **Factors**

Basic Mathematical Setup

Problem Setup

- Two sources of information: gene expression data and factor data

Problem Setup

- Two sources of information: gene expression data and factor data
- **Objective:** identify genes whose expression levels are associated with factors (disease condition, age, biomarkers)

Problem Setup

- Two sources of information: gene expression data and factor data
- **Objective:** identify genes whose expression levels are associated with factors (disease condition, age, biomarkers)
- Gene expression data (e.g. microarray, log scale) is a genes $\times$ samples matrix Y :

Y_{ji} = (log) expression of the j -th gene in the i -th sample

Problem Setup

- Two sources of information: gene expression data and factor data
- **Objective:** identify genes whose expression levels are associated with factors (disease condition, age, biomarkers)
- Gene expression data (e.g. microarray, log scale) is a genes $\times$ samples matrix Y :

Y_{ji} = (log) expression of the j -th gene in the i -th sample

- Expression of the j -th gene across all N samples:

$$Y_j = (Y_{j1} \ Y_{j2} \ \dots \ Y_{jN})^T$$

The Design Matrix

- Factor data is stored in a **full-rank design matrix** $X \in \mathbb{R}^{N \times m}$

The Design Matrix

- Factor data is stored in a **full-rank design matrix** $X \in \mathbb{R}^{N \times m}$
- Each column of X is either
 - a **covariate**: age, weight, molecular phenotypes, biomarkers, or
 - a **factor**: groups (healthy vs disease) or treatments

The Design Matrix

- Factor data is stored in a **full-rank design matrix** $X \in \mathbb{R}^{N \times m}$
- Each column of X is either
 - a **covariate**: age, weight, molecular phenotypes, biomarkers, or
 - a **factor**: groups (healthy vs disease) or treatments

Continuous covariates

$$X = \begin{pmatrix} 23 & 80 \\ 24 & 63 \\ 25 & 92 \\ 26 & 84 \\ 27 & 69 \end{pmatrix}$$

Columns: Age (years), Weight (kg)

Categorical factor

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

Columns: Healthy, Disease

Linear Models (Microarray Data)

- Linear model between Y_j (or normalized Y_j) and X :

$$Y_j = X\beta_j + \varepsilon_j$$

- Linear model between Y_j (or normalized Y_j) and X :

$$Y_j = X\beta_j + \varepsilon_j$$

- $\beta_j \in \mathbb{R}^m$ is an m -dimensional coefficient vector for gene j

- Linear model between Y_j (or normalized Y_j) and X :

$$Y_j = X\beta_j + \varepsilon_j$$

- $\beta_j \in \mathbb{R}^m$ is an m -dimensional coefficient vector for gene j

Interpretation of the coefficients

One interpretation of β_{jk} : the **expected level (or value)** of gene j in the k -th factor (covariate).

- **Least squares**, when all samples have the same error variance

- **Least squares**, when all samples have the same error variance
- **Weighted least squares**, if

$$\text{Var}(Y_j) = W_j \sigma_j^2$$

- σ_j^2 : error variance associated with the j -th gene
- W_j : positive definite weight matrix

Estimation of β_j

- **Least squares**, when all samples have the same error variance
- **Weighted least squares**, if

$$\text{Var}(Y_j) = W_j \sigma_j^2$$

- ▶ σ_j^2 : error variance associated with the j -th gene
- ▶ W_j : positive definite weight matrix
- Estimated value:

$$\hat{\beta}_j = (X^T W_j^{-1} X)^{-1} X^T W_j^{-1} Y_j$$

(with $W_j = I$ this reduces to ordinary least squares, $\hat{\beta}_j = (X^T X)^{-1} X^T Y_j$)

- The covariance of the estimated coefficients:

$$\text{Cov}(\hat{\beta}_j) = s_j^2 V_j, \quad V_j = (X^T W_j^{-1} X)^{-1}$$

Covariance of the Estimates

- The covariance of the estimated coefficients:

$$\text{Cov}(\hat{\beta}_j) = s_j^2 V_j, \quad V_j = (X^T W_j^{-1} X)^{-1}$$

- s_j^2 is the estimate of the error variance σ_j^2 :

$$s_j^2 = \frac{1}{N - m} (Y_j - X \hat{\beta}_j)^T W_j^{-1} (Y_j - X \hat{\beta}_j)$$

- Often we are interested in **combinations** of the $\hat{\beta}_j$, e.g. differences between groups

Contrast Matrix

- Often we are interested in **combinations** of the $\hat{\beta}_j$, e.g. differences between groups
- Three groups: Healthy (H), Moderate (M), Disease (D). Each column of C is one contrast:

C	H - M	M - D	D - H
β_{j1} (H)	1	0	-1
β_{j2} (M)	-1	1	0
β_{j3} (D)	0	-1	1

$$\tilde{\beta}_j = C^T \hat{\beta}_j = \begin{pmatrix} \hat{\beta}_{j1} - \hat{\beta}_{j2} \\ \hat{\beta}_{j2} - \hat{\beta}_{j3} \\ \hat{\beta}_{j3} - \hat{\beta}_{j1} \end{pmatrix}$$

$$\text{Cov}(\tilde{\beta}_j) = s_j^2 C^T V_j C$$

Testing Coefficients Separately

- Each coefficient of $\tilde{\beta}_j$ is tested separately, $H_0 : \tilde{\beta}_{jk} = 0$

Testing Coefficients Separately

- Each coefficient of $\tilde{\beta}_j$ is tested separately, $H_0 : \tilde{\beta}_{jk} = 0$

$$t_{jk} = \frac{\tilde{\beta}_{jk}}{s_j \sqrt{v_{jk}}}$$

- ▶ v_{jk} : k -th diagonal element of $C^T V_j C$ (of V_j when testing $\hat{\beta}_j$ directly)

Testing Coefficients Separately

- Each coefficient of $\tilde{\beta}_j$ is tested separately, $H_0 : \tilde{\beta}_{jk} = 0$

$$t_{jk} = \frac{\tilde{\beta}_{jk}}{s_j \sqrt{v_{jk}}}$$

- ▶ v_{jk} : k -th diagonal element of $C^T V_j C$ (of V_j when testing $\hat{\beta}_j$ directly)
- Under H_0 , t_{jk} follows a **t-distribution** with degrees of freedom

$$d_j = N - \text{rank}(X) = N - m$$

Two Groups: The Two-Sample t-Test

- For two factors (groups), the t-test above is **equivalent to the two-sample t-test**

Two Groups: The Two-Sample t-Test

- For two factors (groups), the t-test above is **equivalent to the two-sample t-test**
- With $X = [\text{Healthy}, \text{Disease}]$ indicators and $W_j = I$:

$$\hat{\beta}_j = \begin{pmatrix} \bar{Y}_H \\ \bar{Y}_D \end{pmatrix}, \quad V_j = \begin{pmatrix} 1/n_H & 0 \\ 0 & 1/n_D \end{pmatrix}$$

Two Groups: The Two-Sample t-Test

- For two factors (groups), the t-test above is **equivalent to the two-sample t-test**
- With $X = [\text{Healthy}, \text{Disease}]$ indicators and $W_j = I$:

$$\hat{\beta}_j = \begin{pmatrix} \bar{Y}_H \\ \bar{Y}_D \end{pmatrix}, \quad V_j = \begin{pmatrix} 1/n_H & 0 \\ 0 & 1/n_D \end{pmatrix}$$

- The contrast $c = (1, -1)^T$ gives

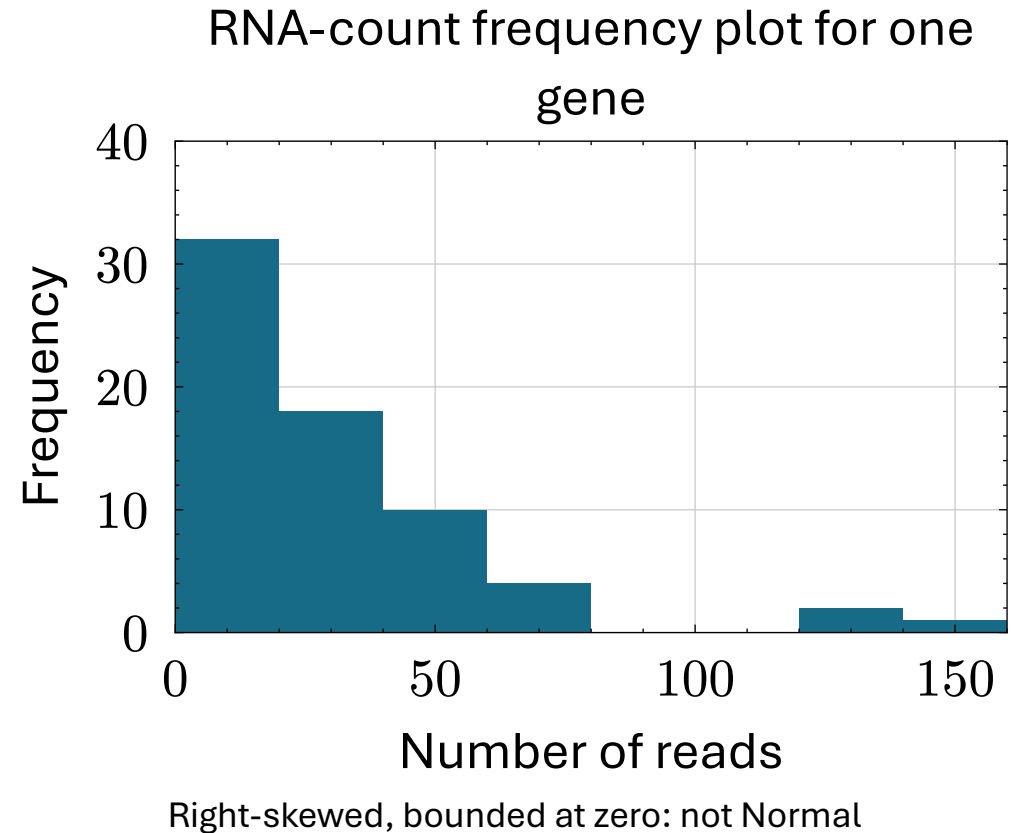
$$t_j = \frac{\bar{Y}_H - \bar{Y}_D}{s_j \sqrt{1/n_H + 1/n_D}}, \quad d_j = N - 2$$

which is exactly the pooled-variance two-sample t-statistic.

RNA-seq Data: Poisson Model

Why Not OLS for RNA-seq?

- RNA-seq produces **count data**
 - Non-negative integers
 - Do not follow a Normal distribution
- Hence we **cannot** use the linear regression model (OLS) as done earlier



The Poisson Model

- Model the count Y_j for the j -th gene with a Poisson distribution, $Y_j \sim \text{Poisson}(\lambda_j)$:

Possion Distribution

$$P(Y_j = Y_{ij}; \lambda) = \frac{\lambda_{ij}^{Y_{ij}} e^{-\lambda}}{Y_{ij}!}$$

The Poisson Model

- Model the count Y_j for the j -th gene with a Poisson distribution, $Y_j \sim \text{Poisson}(\lambda_j)$:

Possion Distribution

$$P(Y_j = Y_{ij}; \lambda) = \frac{\lambda_{ij}^{Y_{ij}} e^{-\lambda}}{Y_{ij}!}$$

- λ_j : average number of reads

The Poisson Model

- Model the count Y_j for the j -th gene with a Poisson distribution, $Y_j \sim \text{Poisson}(\lambda_j)$:

Possion Distribution

$$P(Y_j = Y_{ij}; \lambda) = \frac{\lambda_{ij}^{Y_{ij}} e^{-\lambda_{ij}}}{Y_{ij}!}$$

- λ_j : average number of reads
- Mean and variance are equal:

$$E[Y_j] = \text{Var}[Y_j] = \lambda_j$$

Notation from here on: Y_{ij} = count of gene j in sample i .

Poisson Regression

- Experiment with m design variables $x = (x_1 \ x_2 \ \dots \ x_m)^T$

Poisson Regression

- Experiment with m design variables $x = (x_1 \ x_2 \ \dots \ x_m)^T$
- Model between λ_j and x :

$$\log \lambda_j = \beta_{0j} + \beta_{1j}x_1 + \beta_{2j}x_2 + \dots + \beta_{mj}x_m = \beta_{0j} + x^T \beta_j$$

$$\text{where } \beta_j = (\beta_{1j} \ \beta_{2j} \ \dots \ \beta_{mj})^T$$

Poisson Regression

- Experiment with m design variables $x = (x_1 \ x_2 \ \dots \ x_m)^T$
- Model between λ_j and x :

$$\log \lambda_j = \beta_{0j} + \beta_{1j}x_1 + \beta_{2j}x_2 + \dots + \beta_{mj}x_m = \beta_{0j} + x^T \beta_j$$

$$\text{where } \beta_j = (\beta_{1j} \ \beta_{2j} \ \dots \ \beta_{mj})^T$$

? Question

Why model $\log \lambda_j$ instead of λ_j ?

Poisson Regression

- Experiment with m design variables $x = (x_1 \ x_2 \ \dots \ x_m)^T$
- Model between λ_j and x :

$$\log \lambda_j = \beta_{0j} + \beta_{1j}x_1 + \beta_{2j}x_2 + \dots + \beta_{mj}x_m = \beta_{0j} + x^T \beta_j$$

$$\text{where } \beta_j = (\beta_{1j} \ \beta_{2j} \ \dots \ \beta_{mj})^T$$

? Question

Why model $\log \lambda_j$ instead of λ_j ?



Answer

$\lambda_j > 0$, but $\beta_{0j} + x^T \beta_j$ can be any real number. The log maps $(0, \infty)$ onto $\mathbb{R}$.

Example: A Three-Group Study

- Groups: Healthy (H), Moderate (M), Disease (D)

$$\beta_j = \begin{pmatrix} \beta_{Hj} \\ \beta_{Mj} \\ \beta_{Dj} \end{pmatrix}, \quad x = \begin{pmatrix} x_H \\ x_M \\ x_D \end{pmatrix}$$

Example: A Three-Group Study

- Groups: Healthy (H), Moderate (M), Disease (D)

$$\beta_j = \begin{pmatrix} \beta_{Hj} \\ \beta_{Mj} \\ \beta_{Dj} \end{pmatrix}, \quad x = \begin{pmatrix} x_H \\ x_M \\ x_D \end{pmatrix}$$

- Study with 8 individuals:

Sample i	1	2	3	4	5	6	7	8
Y_{ij}	100	80	30	40	60	0	10	40
x_H	0	0	0	1	1	0	0	1
x_M	0	1	0	0	0	1	0	0
x_D	1	0	1	0	0	0	1	0

The three indicator rows form the columns of the 8×3 design matrix X .

The Model for the Example

- Then, the model for gene j in sample i :

$$Y_{ij} \sim \text{Poisson}(\lambda_{ij}) = \text{Poisson}(e^{\beta_{0j} + x_i^T \beta_j})$$

The Model for the Example

- Then, the model for gene j in sample i :

$$Y_{ij} \sim \text{Poisson}(\lambda_{ij}) = \text{Poisson}(e^{\beta_{0j} + x_i^T \beta_j})$$

Identifiability

With an intercept β_{0j} **and** all three indicators, X is not full rank, since $x_H + x_M + x_D = 1$ for every sample. In practice one indicator is dropped (a reference level, e.g. Healthy); β_{Mj} and β_{Dj} then become log fold changes relative to Healthy.

- To establish the relationship between Y_j and the x variables, we need to **fit a model**

- To establish the relationship between Y_j and the x variables, we need to **fit a model**
- Construct the likelihood function (or the log-likelihood):

$$\mathcal{L}(\beta_{0j}, \beta_j \mid Y_j, X) = \prod_{i=1}^N P(Y = Y_{ij} \mid x_i) = \prod_{i=1}^N \frac{\lambda_{ij}^{Y_{ij}} e^{-\lambda_{ij}}}{Y_{ij}!}$$

- To establish the relationship between Y_j and the x variables, we need to **fit a model**
- Construct the likelihood function (or the log-likelihood):

$$\mathcal{L}(\beta_{0j}, \beta_j \mid Y_j, X) = \prod_{i=1}^N P(Y = Y_{ij} \mid x_i) = \prod_{i=1}^N \frac{\lambda_{ij}^{Y_{ij}} e^{-\lambda_{ij}}}{Y_{ij}!}$$

- where $\lambda_{ij} = e^{\beta_{0j} + x_i^T \beta_j}$ and β_j is a vector

$$\ell = \sum_{i=1}^N \left[Y_{ij} \log \lambda_{ij} - \lambda_{ij} - \log Y_{(ij)}! \right] = \sum_{i=1}^N \left[Y_{ij} (\beta_{0j} + x_i^T \beta_j) - e^{\beta_{0j} + x_i^T \beta_j} \right] + \text{const}$$

$$\ell = \sum_{i=1}^N \left[Y_{ij} \log \lambda_{ij} - \lambda_{ij} - \log Y_{(ij)!} \right] = \sum_{i=1}^N \left[Y_{ij} (\beta_{0j} + x_i^T \beta_j) - e^{\beta_{0j} + x_i^T \beta_j} \right] + \text{const}$$

- Set the derivatives to zero:

$$\frac{\partial \ell}{\partial \beta_{0j}} = \sum_{i=1}^N (Y_{ij} - \lambda_{ij}) = 0, \quad \frac{\partial \ell}{\partial \beta_{kj}} = \sum_{i=1}^N x_{ik} (Y_{ij} - \lambda_{ij}) = 0$$

$$\ell = \sum_{i=1}^N \left[Y_{ij} \log \lambda_{ij} - \lambda_{ij} - \log Y_{(ij)}! \right] = \sum_{i=1}^N \left[Y_{ij} (\beta_{0j} + x_i^T \beta_j) - e^{\beta_{0j} + x_i^T \beta_j} \right] + \text{const}$$

- Set the derivatives to zero:

$$\frac{\partial \ell}{\partial \beta_{0j}} = \sum_{i=1}^N (Y_{ij} - \lambda_{ij}) = 0, \quad \frac{\partial \ell}{\partial \beta_{kj}} = \sum_{i=1}^N x_{ik} (Y_{ij} - \lambda_{ij}) = 0$$

- Solving (numerically, e.g. Newton–Raphson / IRLS) gives $\hat{\beta}_{0j}, \hat{\beta}_{1j}, \dots, \hat{\beta}_{mj}$

Overdispersion

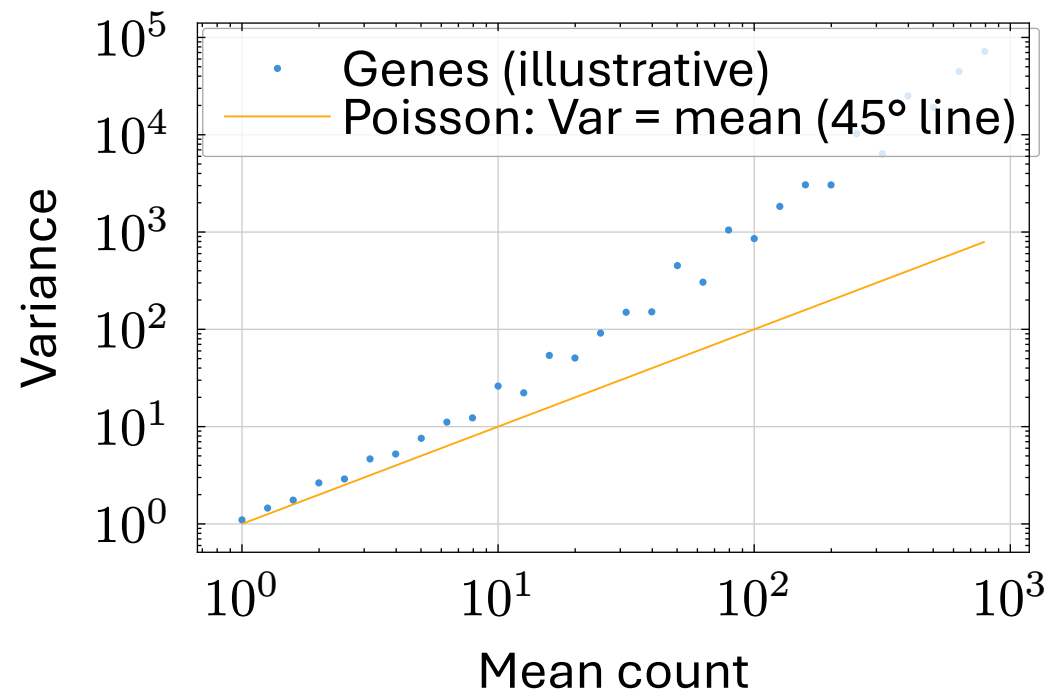
- Poisson distribution:

$$E[Y_{ij} | x_i] = \text{Var}[Y_{ij} | x_i] = \lambda_{ij}$$

- RNA-seq data:

$$\text{Var}[Y_{ij} | x_i] > E[Y_{ij} | x_i]$$

- Genes with higher mean count have higher variance
- The Poisson model **underestimates** the variance



Negative Binomial Regression

The Negative Binomial Model

$$Y_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

The Negative Binomial Model

$$Y_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

$$E[Y_{ij}] = \mu_{ij}, \quad \text{Var}(Y_{ij}) = \mu_{ij}(1 + \alpha_j \mu_{ij}) = \mu_{ij} + \alpha_j \mu_{ij}^2$$

The Negative Binomial Model

$$Y_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

$$E[Y_{ij}] = \mu_{ij}, \quad \text{Var}(Y_{ij}) = \mu_{ij}(1 + \alpha_j\mu_{ij}) = \mu_{ij} + \alpha_j\mu_{ij}^2$$

- α_j : **over-dispersion parameter**, gene-specific

The Negative Binomial Model

$$Y_{ij} \sim \text{NB}(\mu_{ij}, \alpha_j)$$

$$E[Y_{ij}] = \mu_{ij}, \quad \text{Var}(Y_{ij}) = \mu_{ij}(1 + \alpha_j\mu_{ij}) = \mu_{ij} + \alpha_j\mu_{ij}^2$$

- α_j : **over-dispersion parameter**, gene-specific
- $\alpha_j = 0 \implies$ the NB distribution reduces to the **Poisson** distribution

Negative Binomial Probability Mass Function

- Writing $\sigma_{ij}^2 = \mu_{ij}(1 + \alpha_j \mu_{ij})$ for the variance:

$$\Pr(Y_j = Y_{ij}) = \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}, \quad r_{ij} = \frac{\mu_{ij}^2}{\sigma_{ij}^2 - \mu_{ij}}$$

Negative Binomial Probability Mass Function

- Writing $\sigma_{ij}^2 = \mu_{ij}(1 + \alpha_j \mu_{ij})$ for the variance:

$$\Pr(Y_j = Y_{ij}) = \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}, \quad r_{ij} = \frac{\mu_{ij}^2}{\sigma_{ij}^2 - \mu_{ij}}$$

- Since $\sigma_{ij}^2 - \mu_{ij} = \alpha_j \mu_{ij}^2$, we get $r_{ij} = 1/\alpha_j$, i.e.

$$\Pr(Y_j = Y_{ij}) = \binom{Y_{ij} + 1/\alpha_j - 1}{Y_{ij}} \left(\frac{\alpha_j \mu_{ij}}{1 + \alpha_j \mu_{ij}}\right)^{Y_{ij}} \left(\frac{1}{1 + \alpha_j \mu_{ij}}\right)^{1/\alpha_j}$$

The binomial coefficient with non-integer argument is understood via the Gamma function: $\Gamma(y + r)/(\Gamma(r) y!)$.

Modelling the Mean: Normalization

$$\mu_{ij} = c_{ij} q_{ij}$$

- μ_{ij} : mean count for gene j in the i -th sample
- c_{ij} : scaling factor (normalization)
- q_{ij} : the true concentration of fragments from gene j in sample i

Modelling the Mean: Normalization

$$\mu_{ij} = c_{ij} q_{ij}$$

- μ_{ij} : mean count for gene j in the i -th sample
- c_{ij} : scaling factor (normalization)
- q_{ij} : the true concentration of fragments from gene j in sample i
- Assume $c_{ij} = c_i$: **one constant for all genes** in the i -th sample

Modelling the Mean: Normalization

$$\mu_{ij} = c_{ij} q_{ij}$$

- μ_{ij} : mean count for gene j in the i -th sample
- c_{ij} : scaling factor (normalization)
- q_{ij} : the true concentration of fragments from gene j in sample i
- Assume $c_{ij} = c_i$: **one constant for all genes** in the i -th sample
- Model between q_{ij} and the design variables:

$$\log_2(q_{ij}) = \beta_{0j} + x_i^T \beta_j = \beta_{0j} + x_{i1}\beta_{1j} + x_{i2}\beta_{2j} + \dots$$

Estimation of the Size Factors c_i

- Median-of-ratios estimate for the i -th sample:

$$c_i = \operatorname{median}_{j: Y_j^R \neq 0} \frac{Y_{ij}}{Y_j^R}$$

- Pseudo-reference: geometric mean of gene j across samples

$$Y_j^R = \left(\prod_{i=1}^N Y_{ij} \right)^{1/N}$$

- Genes with a zero count in any sample have $Y_j^R = 0$ and are excluded



Toy example (2 samples)

Gene	Y_{1j}	Y_{2j}	Y_j^R	Ratios
A	10	20	14.14	0.71, 1.41
B	50	100	70.71	0.71, 1.41
C	0	5	0	excluded
D	30	90	51.96	0.58, 1.73

$c_1 = 0.71$, $c_2 = 1.41$: sample 2 is sequenced about twice as deeply.

Likelihood for the NB Model

$$\mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X) = \prod_{i=1}^N \Pr(Y_{ij}) = \prod_{i=1}^N \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}$$

Likelihood for the NB Model

$$\mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X) = \prod_{i=1}^N \Pr(Y_{ij}) = \prod_{i=1}^N \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}$$

- where

$$\mu_{ij} = c_i 2^{\beta_{0j} + x_i^T \beta_j}$$

Likelihood for the NB Model

$$\mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X) = \prod_{i=1}^N \Pr(Y_{ij}) = \prod_{i=1}^N \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}$$

- where

$$\mu_{ij} = c_i 2^{\beta_{0j} + x_i^T \beta_j}$$

- Estimate the parameters β_{0j} , β_j (a vector) and α_j by maximizing $\mathcal{L}$:

$$(\hat{\beta}_{0j}, \hat{\beta}_j, \hat{\alpha}_j) = \arg \max_{\beta_{0j}, \beta_j, \alpha_j} \mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X)$$

Likelihood for the NB Model

$$\mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X) = \prod_{i=1}^N \Pr(Y_{ij}) = \prod_{i=1}^N \binom{Y_{ij} + r_{ij} - 1}{Y_{ij}} \left(1 - \frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{Y_{ij}} \left(\frac{\mu_{ij}}{\sigma_{ij}^2}\right)^{r_{ij}}$$

- where

$$\mu_{ij} = c_i 2^{\beta_{0j} + x_i^T \beta_j}$$

- Estimate the parameters β_{0j} , β_j (a vector) and α_j by maximizing $\mathcal{L}$:

$$(\hat{\beta}_{0j}, \hat{\beta}_j, \hat{\alpha}_j) = \arg \max_{\beta_{0j}, \beta_j, \alpha_j} \mathcal{L}(\beta_{0j}, \beta_j, \alpha_j \mid Y_j, X)$$

- Then perform **Wald test(s)** on the $\hat{\beta}_j$ values or on contrasts $\tilde{\beta}_j$

Hypothesis Testing: The Wald Test

- For a coefficient of β_j :

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0 \quad \text{or} \quad H_0 : \beta_j = \theta \quad \text{vs} \quad H_1 : \beta_j \neq \theta$$

- For a coefficient of β_j :

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0 \quad \text{or} \quad H_0 : \beta_j = \theta \quad \text{vs} \quad H_1 : \beta_j \neq \theta$$

- Wald statistic:

$$W = \frac{(\hat{\beta}_j - \theta)^2}{\text{Var}(\hat{\beta}_j)}$$

The Wald Test

- For a coefficient of β_j :

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0 \quad \text{or} \quad H_0 : \beta_j = \theta \quad \text{vs} \quad H_1 : \beta_j \neq \theta$$

- Wald statistic:

$$W = \frac{(\hat{\beta}_j - \theta)^2}{\text{Var}(\hat{\beta}_j)}$$

- Under H_0 , $W \sim \chi^2$ -distribution with **1 degree of freedom**

The Wald Statistic in DESeq2

- DESeq2 uses $\sqrt{W}$ as the statistic:

$$W_s = \sqrt{W} = \frac{\hat{\beta}_j - \theta}{\text{SE}(\hat{\beta}_j)}$$

- W_s is approximately standard Normal for large samples; it follows a t-distribution only under several strong assumptions

The Wald Statistic in DESeq2

- DESeq2 uses $\sqrt{W}$ as the statistic:

$$W_s = \sqrt{W} = \frac{\hat{\beta}_j - \theta}{\text{SE}(\hat{\beta}_j)}$$

- W_s is approximately standard Normal for large samples; it follows a t-distribution only under several strong assumptions

i Several parameters at once

The Wald test can also compare multiple parameters at the same time. For q contrasts $C^T \beta_j$:

$$W = (C^T \hat{\beta}_j - \theta)^T (C^T \Sigma_j C)^{-1} (C^T \hat{\beta}_j - \theta) \sim \chi_q^2, \quad \Sigma_j = \text{Cov}(\hat{\beta}_j)$$

Summary

Microarray vs RNA-seq

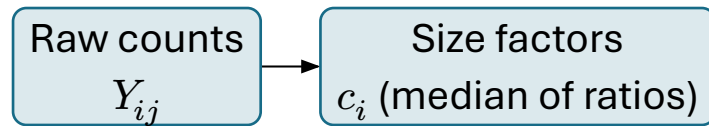
	Microarray	RNA-seq
Data	Continuous (log) intensities	Non-negative integer counts
Distribution	Normal	Negative Binomial ($\alpha_j = 0$: Poisson)
Model	$Y_j = X\beta_j + \varepsilon_j$	$\log_2 q_{ij} = \beta_{0j} + x_i^T \beta_j$
Normalization	Before fitting	Size factors c_i in the mean
Estimation	(Weighted) least squares	Maximum likelihood (numerical)
Test	t-test, $d_j = N - m$	Wald test, $W_s = \hat{\beta}_j / \text{SE}$

The DESeq2-Style Workflow

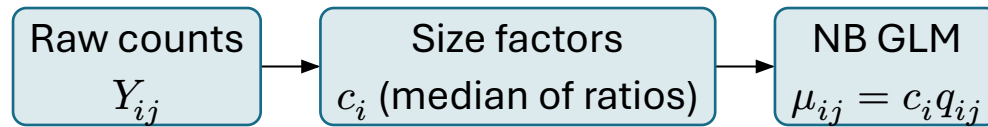
Raw counts

Y_{ij}

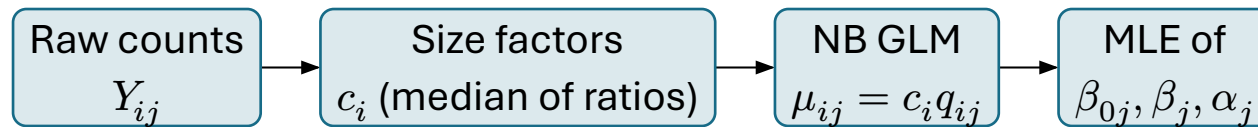
The DESeq2-Style Workflow



The DESeq2-Style Workflow



The DESeq2-Style Workflow



The DESeq2-Style Workflow

